Edgar Omar Reyes Páiz

Software Developer · Software Engineer · C# / Unity Specialist

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PROFESSIONAL SUMMARY

Software Developer with **10+ years of professional experience** designing, developing, testing and deploying software across desktop, mobile, interactive, XR and hardware-integrated environments. Strong background in **C#, software architecture, modular system design and real-time applications**, complemented by experience with C/C++, JavaScript, Python, Bash and Linux.

Experienced across requirements analysis, architecture, implementation, debugging, testing, integration, deployment and production troubleshooting. Currently working through **xrlab.dev** on custom software solutions and interactive systems, while expanding into broader software engineering, systems, automation and technology roles.

CORE SKILLS

Software Engineering
Software architecture · OOP · SOLID · Clean Code · Modular design · Event-driven systems · System programming · Tool development · Debugging · Testing · API / SDK integration · Git / GitHub

Programming
C# (advanced / primary) · C / C++ · JavaScript · Bash · Python · Java

Systems & Linux
Linux · Arch Linux · Fedora · Debian · Xorg · Wayland · Terminal workflows · System configuration · Troubleshooting · Server environments

Unity & Real-Time Development
Unity 2022 LTS / Unity 6 · Unity C# · uGUI · UI Toolkit · ScriptableObjects · Custom tools and plugins · Gameplay · Interactive applications · Performance optimization

XR & AI
AR / VR / XR · WebXR · AR Foundation · ARKit · ARCore · Vuforia · SteamVR · OpenAI · ComfyUI · Qwen · FLUX · Local AI models

Integration & Multimedia
Hardware/software integration · Arduino · APIs · SDKs · SQL · Firebase · Webcams · QR workflows · FFmpeg · ffprobe · ffplay · RenderTexture

PROFESSIONAL EXPERIENCE

xrlab.dev

Independent Software Developer / Technical Lead | 2023–Present | Mexico

- Design, develop and deploy custom software solutions, interactive applications and real-time systems using Unity, C# and supporting technologies.
- Translate business and technical requirements into practical, modular software architectures using SOLID principles and separation of concerns.
- Develop applications for desktop systems, kiosks, touchscreens, events and dedicated hardware, including offline-first deployments.
- Integrate APIs, external services, cameras, peripherals, QR workflows and hardware/software interfaces.
- Implement and evaluate AI-powered workflows using OpenAI and local image-generation technologies, including ComfyUI, Qwen and FLUX.
- Perform debugging, functional testing, hardware validation, deployment and production troubleshooting.
- Manage projects from requirements analysis and technical planning through implementation and delivery.

Selected work: BuildYourOwnSandwich · Interactive3DViewer · NarraGol · Klar Photobooth · xrlab AI / Interactive AI Platform

Dynamic Media Triad

Unity Developer | August 2018–October 2023 | Remote

- Developed core systems and gameplay functionality for professional game projects including **Astrodogs** and **BR2000**.
- Implemented UI/UX systems and integrated and optimized production assets across multiple platforms.
- Worked with the Nintendo SDK and API for Unity and contributed to platform-specific development and optimization.
- Managed GitHub repositories and collaborative development workflows.
- Worked across system programming, gameplay programming, debugging and application integration.

Freelance Developer

August 2018–Present

- Developed software, games, XR applications and interactive experiences using Unity and C#.
- Provided technical consulting for Unity and Unreal Engine projects, including asset integration, optimization and development workflows.
- Supported project planning, estimation and technical decision-making.
- Developed interactive effects and node-based filters using TikTok Effect House and Meta Spark Studio.
- Worked directly with clients to define requirements and deliver technical solutions.

EON Reality Mexico

Unity C# Developer | April 2018–August 2018 | Mexico

- Developed VR gameplay and networking systems using Unity and C#.
- Implemented UI, navigation and UX functionality and integrated 3D and multimedia assets.
- Developed Unity-to-Arduino communication and a custom Unity plugin for Arduino integration.
- Tested and validated hardware/software interactions.

Lightsweep / Cinetransformer

Unity XR Developer | February 2017–February 2018 | Mexico

- Developed AR/VR applications for HTC Vive, Oculus Rift, Samsung Gear VR and Google Cardboard.
- Developed Unity applications for Android and iOS using SteamVR, Vuforia, ARKit, ARCore and AR Foundation.
- Designed UX/navigation and integrated and optimized 3D assets from Maya.
- Integrated SQL databases with Unity and worked with photography, video and audio production.
- Tested Unreal Engine VR applications.

WashaWasha Games

Unity Programmer | February 2016–August 2016 | Mexico

- Developed Unity systems and tools for level and game designers.
- Created custom tools to improve development workflows.
- Developed gameplay functionality for *Learn Math with El Chavo* and integrated audio and interactive gameplay systems.

EDUCATION

Middlesex University
BA Interactive Animation — First Class Honours | 2013–2016

SAE Institute Mexico
Licenciatura en Diseño de Videojuegos | 2011–2015

ADDITIONAL TECHNOLOGY

Development: .NET · Unreal Engine · Git · GitHub
Platforms: Linux · Windows · Android · iOS · Nintendo
Languages: Spanish — Native · English — Fluent